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Cryptographic Defenses for Machine Learning Privacy in IoT Systems

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Abstract
The rapid expansion of computational and networking capabilities has established the Internet of Things (IoT) as a cornerstone of modern real-time systems. As IoT devices proliferate, they generate vast streams of data—much of it sensitive—necessitating robust privacy protection mechanisms to prevent unauthorized access and misuse. While numerous machine learning models have been proposed to address these privacy concerns, many still fall short in terms of efficiency, scalability, or ease of deployment in resource-constrained IoT environments. This paper introduces the Advanced Privacy Protection (APP) Machine Learning Model, a novel architecture designed to enhance both the confidentiality and utility of IoT data analytics. By integrating advanced cryptographic techniques, the model ensures that sensitive information remains secure throughout the data lifecycle—from collection and storage to processing and analysis. A core innovation of APP is the Secure Data Provider (SDP), which orchestrates privacy-preserving training directly on distributed nodes, minimizing the risk of data exposure. For supervised learning tasks, the model employs Support Vector Machines (SVM), chosen for their balance of performance and interpretability in heterogeneous IoT datasets. Empirical evaluation demonstrates that the proposed model not only achieves higher classification accuracy compared to existing privacy-preserving approaches but also reduces communication overhead and computational complexity. These improvements make the system practical for deployment across diverse IoT ecosystems, where both data utility and privacy are paramount. By unifying cryptographic security with efficient machine learning, the APP framework offers a scalable solution for safeguarding sensitive information in the era of pervasive connectivity.

Article Highlights

- The proposed article will provide a well-trained secure model for a secure data provider using the encrypted data of devices.
- Our article ensures the cryptography-based security in cloud-server domains, which offer a substantial amount of computational and storage capacity.
- Here the data providers employ secret sharing technologies or homomorphic encryption.

Keywords Advanced privacy protection (APP) · Secure data provider (SDP) · Support vector machine (SVM)

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1 Introduction

Surveys indicate that the IoT has experienced a volatile increase in data in recent years. The ability of continuous surveillance enables the IoT to significantly alter both the way people live and the way they work [1]. The term "IoT" refers to ubiquitous devices, sensor networks, and intelligent sensor objects that exchange data and provide services via sensor networks; these components are essential for the development of smart environments [2]. The proliferation of IoT-enabled devices at an exponential rate has garnered the interest of both the academic and industrial communities. The continuous advancements of the Internet of Things have greatly expanded the range of connectivity between remote devices that are connected to the internet in order to transfer data and access [3]. In conjunction with sensors, these devices not only acquire data in real-time but also facilitate prompt and efficient transmission of the collected information to the relevant stakeholders for enforcement of required measures [4].

In order to mitigate these security risk factors, the majority of systems encrypt responsive data with symmetric or asymmetric key mechanisms [5]. Data integrity and authenticity are crucial for data security, as they ensure that the data are as they were obtained by the sensing devices and have not been altered; they are also crucial for dependable IoT applications. These developments improved the efficacy of cryptographic methods for detecting and validating clients and storage servers [6]. This is the method by which an IoT data system authenticates the user and specifies the policies that prevent unauthorized users from accessing the resources. After mutual authentication between the user and an IoT sensor is complete, they are required to establish a Session Key (SK). Through the utilization of the established SK, they are capable of securely exchanging information in order to obtain immediate data access [7].

Despite the potential for cloud systems to offer substantial advantages in the management of vast amounts of data, concerns regarding security and privacy remain, such as the confidentiality of such information [8]. A high probability exists that the data could be altered or removed in the event of a cloud server corruption, thereby compromising the confidentiality and integrity of the data [9]. Users should therefore encrypt their data prior to outsourcing at all times. The cloud system supports a variety of encryption techniques, each with its own advantages and disadvantages, making it difficult to select the most appropriate one, particularly when ensuring the confidentiality, availability, and integrity of data [10].

The proposed model in this paper aims to train a secure model for a secure data provider using the encrypted data of devices. Training in secure outsourcing focuses primarily on cloud computing. Cloud servers offer a substantial amount of computational and storage capacity. To convert their servers, data providers employ secret sharing technologies or homomorphic encryption. Among the major issues facing the Internet of Things (IoT) are privacy and security. Among the difficulties confronting IoT are improper device upgrades, a lack of effective and reliable security procedures, user ignorance, and well-known active device monitoring. This work works on identifying: i. various privacy and security concerns ii. Methods for securing IoT-based environments and systems' constituent parts iii. Current security measures iv. Optimal privacy models required and appropriate for various IoT-driven application layers.

As a result, we present a practicable and efficient framework for training machine learning in an IoT environment that protects privacy. In particular, the exponential growth of data places a significant storage strain on Internet of Things devices that have constrained resources. Hence, the implementation of IoT devices in closer proximity to one another in higher configurations has emerged as a viable resolution. The accumulated information will be stored by IoT devices in adjacent locations. An SDP is presumed to be interested in training its secure model using the data contained in the nodes. During the training procedure, node data will not be compromised and made accessible to SDP and other devices. There will be no disclosure of the SDP model to nodes. As follows, contributions have been made:

(1) Advanced Privacy Protection (APP) is proposed by the characteristics of nodes and the data privacy requirements of the Internet of Things (IoT). Under this approach, nodes are capable of safeguarding the confidentiality of the data while allowing an untrusted SDP to train machine learning models.
(2) The APP employs secure data processing mechanisms. In addition, two secure methods for normalizing data are suggested; these methods can standardize the data distribution across all fog nodes and generate training data of superior quality.

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(3) The ciphertext is referred to as the encrypted data as well and its accuracy is measured in APP, thereby preventing the error that could result from ciphertext computations with inconsistent precision.
(4) The security evaluation demonstrates that the framework satisfies all participants' stringent privacy protection requirements.

The work is organized as follows: Sect. 2 describes the works related to data security in IoT and other communication models. The proposed model and computations are explained with diagrammatic descriptions in Sect. 3. The evaluations and comparisons with evidences are given in Sect. 4. And, the paper is concluded in Sect. 5 with some directions for future work.

2 Related works

A Hashed Needham Schroeder (HNS) Costs Optimised Deep Machine Learning approach was presented in the study [11] to ensure the security of Industrial IoT-DT in a cloud environment. At the outset, HNS Public Key Generation (PKG) computed both the public key and a flag value. Subsequently, the execution time was improved by employing the public key, which restricted data or message exchange over the protected channel to authorized cloud users (CU) who had undergone training. Due to this, the '2' phases were employed to compute the cost function (CF). During the first phase, the CF was computed; in the second, the overall CF was determined; this reduced the computational cost (CC) and communication overhead (CO), which in turn simplified the monitoring and control of the entire process. The primary drawback of this approach was its significantly higher data requirement compared to the alternative methods.

The work [12] proposed secure elongation orthogonal frequency division multiplexing as an alternative transmission system for the Internet of Things to improve spectral efficiency and security while maintaining the efficacy of physical layer outsourcing. The objective was to generate truncated OFDM time domain signals and transfer a portion of them with a randomly selected one using a private matrix. By abbreviating by the offloading power amplifier theorem, a private matrix was generated, which enhanced the power amplifier's offloading efficiency and was highly suitable for use in IoT networks. Furthermore, the proposed framework adhered to the efficient computation algorithm known as Fast Fourier Transforms. However, the STOFDM encountered several significant security vulnerabilities and possessed a greater computational complexity.

Secure outsourcing training involves the delegation of ciphertext data from data proprietors who have restricted computational resources to cloud servers. The machine learning process that ensures privacy is executed by the cloud servers, which also builds a private ML model on behalf of the training service requester. A system for the training and evaluation of privacy-preserving decision trees within a twin-cloud architecture was proposed in the paper [13]. An effective privacy-preserving outsourced SVM method was proposed in the paper [14]. This scheme ensures the confidentiality of both the training data and the SVM model. A secure deep computation paradigm was proposed in the work [15] by offloading the costly operations to the cloud. For multiple parties, the model in [16] proposed an outsourced C4.5 algorithm that protects privacy in horizontally and vertically divided data. The authors [17] put forth a framework for multi-party machine learning that ensures privacy preservation; in this framework, the involvement of data proprietors in the training process is unnecessary. In an outsourced cloud computing environment, the model [18] suggested a clinical decision support system that safeguards confidentiality. Cloud-based machine learning service deployment has been adopted as a training solution. However, this strategy is predicated on the supposition of non-collusion in general, which poses a significant security risk.

Each participant in the secure collaborative training scenario possesses a subset of computation resources in addition to their data. As a result, each participant engages in communication and computation activities while training a global ML model collaboratively using the shared training data. During instruction, data proprietors must maintain the confidentiality of their information [19, 20]. Protocols for addressing the secure multi-party computation problem were proposed in the model [21]. The paper [22] introduced a secure sum protocol that provided robust security guarantees. Two secure gradient-descent methods were proposed utilizing this protocol. FogFL, a fog-enabled federated learning framework proposed by the model [23], aims to reduce energy consumption and communication latency of resource-constrained peripheral devices while facilitating distributed learning. The federated learning scheme proposed in reference [24] is both

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secure and verifiable, enabling the implementation of federated deep learning and ensuring the verifiability of the final learning outcomes. The study [25–29] introduced a non-interactive federated learning (NFL) technique for regression training that ensures privacy protection throughout the entire training procedure. A collaborative architecture was proposed in the model [28, 30–37], which utilized low-orbit satellite network communication and orbital edge computation. The secure system architecture for IoT is presented in the following works.

By designing and implementing a contextual framework, the IoT secure system architecture (SSA) guarantees the security and privacy of IoT devices, data, and interaction. It uses network protocols, hardware, and software to safeguard the IoT system's availability, confidentiality, and integrity [38]. The SSA is made up of essential elements such IoT devices, network infrastructure, cloud services, security protocols, and user interfaces. These elements work together to build the IoT system's foundation and each one adds a unique purpose to the system's overall structure [39]. Sensors, actuators, embedded systems, and other data-gathering and data-transmitting devices are all included in the Internet of Things. In order to link IoT devices and enable data transmission, the network infrastructure offers the underlying communication network. The underlying infrastructure for handling and storing IoT data is provided by cloud servers. Intrusion detection, access control, encryption, and authentication are a few examples of security measures. Device administration, monitoring, and control are made possible by the user interface [40]. The work presented in [41] developed a model for E-Health Monitoring using Artificial Bee Colony Optimization.

3 Proposed model

In this proposed model, the motive of this work is to train the machine learning model for Secure Data Provider (SDP). Simultaneously, the secure data of each node will not be disclosed to SDP and other nodes in cloud. The system model is presented in Fig. 1. The elements that are involved in the proposed model are, IoT elements, fog nodes, SDP and Trusted Party (TP).

3.1 Trusted party (TP)

As a reliable source, TA creates the system parameters that are used by all users.

3.2 IoT elements

An enormous quantity of IoT data will be generated by IoT devices. However, they have very little processing and storage power.

3.3 Fog nodes (FN)

Fog nodes are powerful computers and storage devices. They gather, store, and handle the data produced by Internet of Things devices. Sensitive information includes the data that FNs has, and it cannot be disclosed.

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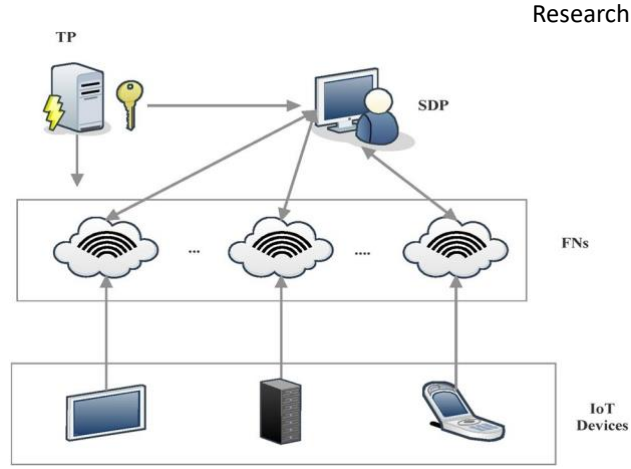

Fig. 1 System Model

3.4 Secure data provider (SDP)

SDP intends to use the information in the FN to train machine learning models. Because SSP is unreliable, it will attempt to acquire FNs' data while doing training.With respect to the system model, the system goal comprises of privacy protection, training the machine learning model, minimal computational and communication overhead.

3.5 Advanced privacy protection:

3.5.1 Training the machine learning model

Ensuring the high accuracy of the trained models is crucial when utilising ciphertext data from FNs to train machine learning models. Therefore, a universal computing framework is framed and uses data normalisation to provide highquality training data.

3.5.2 Low computation and communication overhead

Based on cryptography technology, the proposed approach is aimed to provide privacy-preserving training. Since the technology of cryptography would result in a large amount of processing and communication overhead, the efficiency of the framework is to be maximized.

3.5.3 Threat model

It is assumed in the proposed model that FNs and SDPs are trustworthy yet inquisitive (TP is trusted). While they will honestly carry out the procedure, they attempt to get the information of other participants via the examination of processing outcomes. Simultaneously, we permit a maximum of $(n - 1)$ FNs to conspire with one another to examine the privacy of other respondents and a maximum of $(n - 2)$ FNs to conspire with SDP to examine the privacy of other participants. An opponent is defined to demonstrate the security of the proposed strategy. The attacker can listen in and examine the data during the data transfer. The interactions between participants in the protocol's implementation make up the data transmission process.

3.6 Working process

This section provides a detailed explanation of the proposed framework's operation. APP primarily consists of machine learning model updates, privacy protection machine learning training, data normalization, accuracy detain, and system initialization. Figure 2 depicts the proposed model work flow. It comprises of four sections, IoT data, section for FN parameter generation, section for data encryption and APP model training. Here, the IoT data and generated FN parameters are provided for data encryption in Sect. 3. Further, the data are provided for APP training for privacy protection.

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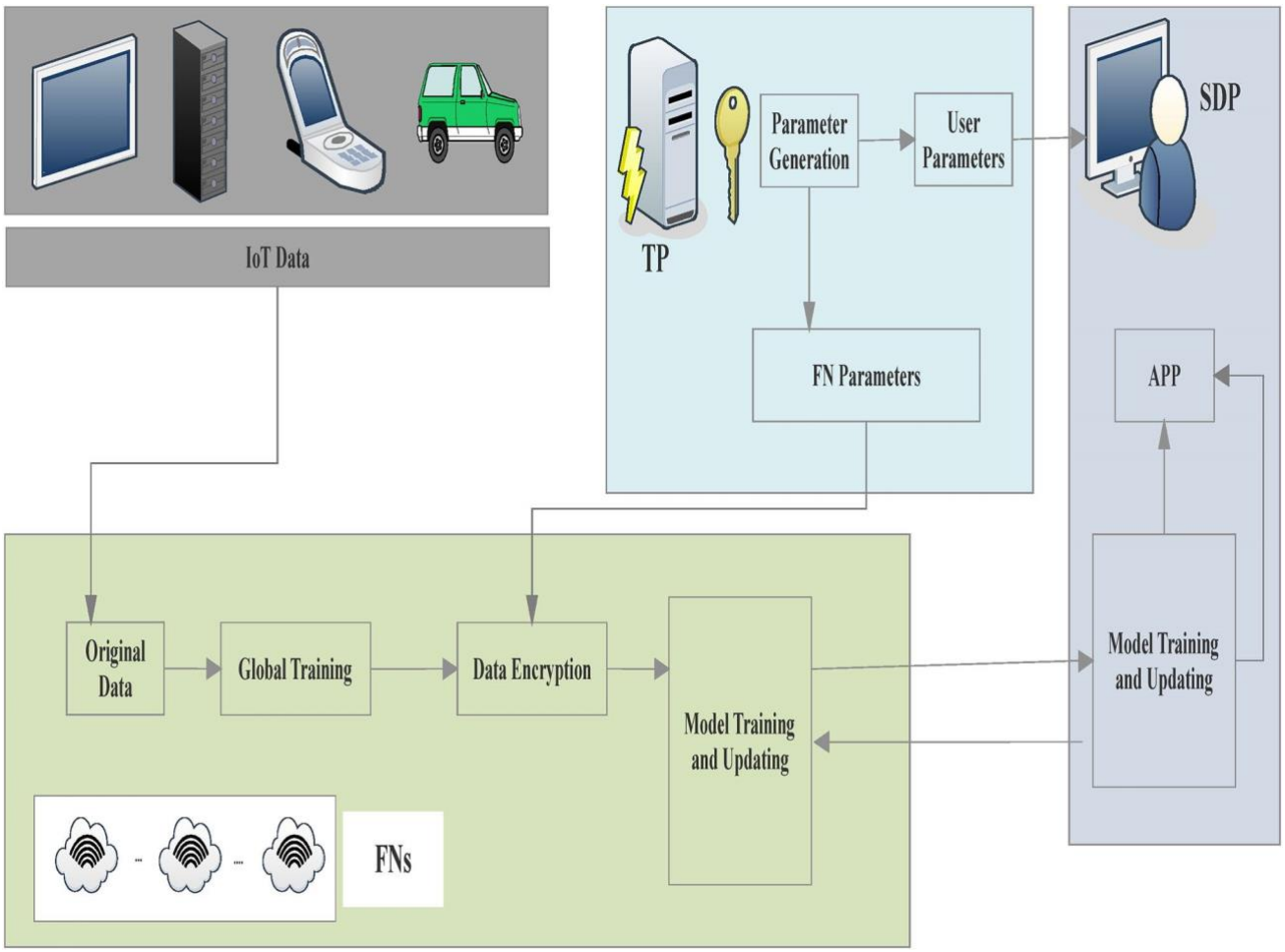

3.6.1 Data normalization

Within the proposed architecture, SDP trains a machine-learning model using the encrypted data of every FN. As a result, normalising the data by all FNs is required and may enhance the calibre of training data. The data format of fog nodes is given as,

$$FN_k = (x_1^{kj}, x_2^{kj}, \ldots, x_d^{kj}, y^{kj}),)j = 1,2, \ldots, n_k, \quad (1)$$

In the process, the participants may not aware about the data of the nodes and the process of protecting z-score is explained below,

i. For i'th dimension of 'd', the FN_k derives,

$$x_i^k = \sum_{j=1}^{n_k} x_i^{kj} \quad (2)$$

Following, encryption operation is performed and the data is sent to SDP along with n_k.

Fig. 2 Workflow of the Proposed Model ii.

Further, SDP derives,

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$$\bar{x}_i = \frac{\sum_{i=1}^{i-1} m_k}{n} \quad (3)$$

The outcome is given to the nodes. iii.

The node computes,

$$\sigma = \sqrt{\frac{\sum_{j=1}^{n_k} (x_i^{kj} - \bar{x}_i)^2}{n_k}} \quad (4)$$

And

$$x_{ikj} = (x_i^{kj} - x_i)/\sigma \quad (5)$$

Further, the computations for min–max are given as follows,

i. The maximum and lowest values of each dimensional data set are calculated by each FN. ii. Following the computation of the global maximum and lowest values, the information will be normalised in accordance with the min–max values and the computation is given in (6).

$$x_i^{kj} = \frac{x_{ikj} - (x_{imin} - A_{mini})}{(x_{imax} + A_{maxi}) - (x_{imin} - A_{mini})} \quad (6)$$

where, 'A' is the rational integer.

3.6.2 Accuracy detain

Many of the data used in the training process for machine learning are floating-point values. For this reason, before encrypting, floating-point numbers must be converted to integers. The original data will be greatly expanded by this procedure. Nevertheless, the encryption algorithm's plaintext space is constrained. The issue of plaintext overflow and data compression will arise if the extended data is used for homomorphic operations. The proposed solution to this issue is Accuracy Detain. The detailed process if explained in the following (Algorithm 1).

For example, in normal conversion method, the floating point number is multiplied by $2^F or 10^F$, where $F \rightarrow accuracyoffloatingpointnumber$, for expanding the data. Because of limitations in encryption algorithm, it can be given as,

$$x2^F.y2^F \rightarrow xy2^F,\ xy2^F + z2^F \rightarrow (xy2^F + z)2^F \neq (xy + z)2^F \quad (7)$$

Using accuracy detain for this, the encrypted data (cip, a), where, cip, is the cipher data and a is the accuracy factor, the expression can be given as,

$$xy.2^{2F} = ([xy.2^{2F}], 2) \quad (8)$$

Algorithm 1 Accuracy detain

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Input: $(x^{t.F}{}_{p_k}, t)$ or $(x^{t.F}{}_{s_k}, t)$ // $p_k \rightarrow public\ key$
, $s_k \rightarrow private\ key$, $i \equiv m$, $F \rightarrow accuracy\ of\ floating\ point\ number$
Output: $([x^F]_{p_k}, 1)$ or $([x^F]_{s_k}, 1)$
Begin
SDP:
i. Send
$([x^{t.F} + A^{t.F}]_{pk_i}, t)$
or $([x^{t.F} A]_{sk_i}, t)$ to FN
$\boldsymbol{FN_k}$**:**
ii. Decrypt
$([x^{t.F} + A^{t.F}]_{pk_i}, t)$
or $([x^{t.F} A]_{sk_i}, t)$
iii. Compute
$x^F + A^F = (x^{t.F} + A^{t.F}$
$x^F A = (x^{t.F} A/2^{(t-1)F}$ and $)/2^{(t-1)F}$ or
Encrypt x^F
iv. Send
$([x^F + A^F]_{p_k}, 1)$ or
$([x^F A]_{sk_i}, 1)$ to SDP
SDP:
v. Compute
$([x^F]_{p_k}, 1) \leftarrow [x^F + A^F]_{p_k}.\ ([x^F A]_{sk_i}, 1).\ [N_k - A^F]_{p_k{}^F}$ or
$([x^F]_{sk_i}, 1) \leftarrow [x^F A]_{sk_i}.A^{-1}$
End

And, the secure addition process is explained in (Algorithm 2).

Algorithm 2 Secure Addition

Input : $g_i, i = 1,2,3, \ldots m$
Output: g
Begin
Nodes:
For $k = 1 \rightarrow m$
i. $\boldsymbol{FN_k}$derives
$g_k{}' = g_k + \beta_k$
ii. Encrypt
$g_k{}'$ with pk_g
iii. Send
$[g_k{}']_{pk_g}$ to SDP
End For
SDP:
Compute
$$[g_k{}']_{pk_g} = \prod_{i=1}^{m} [g_k{}']_{pk_g}$$
iv. Decrypt
$[g_k{}']_{pk_g}$
v. Compute
$g = g' - \beta$
End

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3.7 APP-machine learning model

Here, four training procedures are used in this part using the proposed model for machine learning. Before training, it is expected that the training data has been normalized. Secure SVM training is used here. (Algorithm 3) provides a thorough procedure for training an SVM model. The amount and quality of the total data will be greatly enhanced by the system's ongoing addition of new fog nodes or by the data from existing fog nodes. As a result, updating the learned model of the SDP is crucial. The model update procedure for the model trained using the gradient-descent Method is as follows:

i. TP redistributes the system settings for the fog nodes and SDP that are taking part in the update procedure.
ii. Using the additional data, FNs and SDP train a new model. Here, a new model ' M' ' is obtained for SDP. iii. Based on that, the SDP frames the new model.

Algorithm 3 Secure Training using SVM

Input: Data of Nodes $\boldsymbol{FN_k}$
Learning Rate 'R', Iterations 'P', regularization factor 'T'
Output: Model Factors $M = (m_1, m_2, .., m_d)$
Begin
$\boldsymbol{FN_k}$
For $\boldsymbol{i = 1 \leftarrow n}$
i. $\boldsymbol{FN_k}$encrypts data using p_k
Send $([x^F]_{p_k}, 1)$ and [y] to SDP
End for
SDP:
For $\boldsymbol{i = 1 \leftarrow P}$**:**
2. $([mx^F]_{p_k}, 2) \leftarrow secureproduct([x^F]_{p_k}, m)$
3. $[R1]_{p_k} = [mx^F]_{p_k}.[N_k - 1]_{p_{k0}}(y = 1)$
$[R2]_{p_k} = [mx^F]_{p_k}.[N_k - 1]_{p_{k0}}(y = -1)$
4. Perform secure sign function
5. Perform secure addition
6. To derive $([mx^F]_{p_k}, 2)$, SDP performs secure addition
If$m = 1$, Attack=Yes,
Else if$m = 0$, Attack=No
End if
End if
End For
End

4 Results and discussions

The machine used to build the model has an Inter Core i7 CPU, 16 GB of RAM, and Python 3.8 code. The model applies the classification algorithms FogFL and NFL using Python libraries, Keras, imbalanced learn, and sklearn.

Furthermore, many Internet of Things datasets lack updated versions of novel attack models that have been created recently and have insufficient feature sets. To gather data from various devices, a contemporary data set called IoTID20 [27] is employed in this experiment. Here, IoT intrusion detection for home-based IoT devices is employed for the evaluations and investigation. More than 600,000 samples with 80 characteristics and three different types of labels make up the collection. It includes a vast quantity of flow-based information, including both normal and intrusion activity. The data samples used in this study are from IoTID20; the ID label includes examples of regular data, while the ITI label includes examples of different types of intrusions. In this case, the overall number of normal instances is 15 times lower than the number of intrusion instances. Moreover, the model designed in such a manner that the number of devices increases, the

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data are collected and considered for encryption process. Here, the Table 1 presents the data for experimental environment.

Table 1 Experimental Setup

Operating System	Windows 10
Memory	8G
CPU specifications	Intel(R) Core(TM)i7-10510U, 1.80 GHz, 2.30 GHz
Programming	Python

4.1 Evaluation factors

Classification Accuracy, Sensitivity, Specificity, Error Rate, Precision Rate, and G-Mean Rate are the metrics used to assess the proposed model. Because the distribution of class labels is very uneven and unequal, accuracy has also been given extra weight in order to evaluate the system's efficacy. Therefore, based on all the visual data, the proposed strategy has been shown to be the most reliable way to detect assaults in real-world IoT systems, and it also achieves superior performance across all IoT communication profile classes.

This section discusses the assessment criteria for model analysis in attack detection. True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) constitute the basis for the calculations. The section also contains the formulae for determining the G-Mean, error rate, sensitivity, specificity, and accuracy of classification.

4.1.1 Classification accuracy (CA):

The ratio of accurately identified data to the total number of examples supplied for testing describes the accuracy of the categorization. Additionally, the calculation is

$$CA = \frac{No.\ of\ correctly\ classified\ instances}{No.\ of\ data\ instances} \times 100\% \tag{9}$$

4.1.2 Sensitivity (SN)

This is also known as the recall rate. The following formula may be used to calculate the sensitivity rate based on the True Positive value obtained from the classifications,

$$Sensitivity = \frac{TP}{TP + FN} \tag{10}$$

4.1.3 Specificity rate (SP)

The actual negative rate is the definition of specificity rate, and this is its formula.

$$Specificity = \frac{FP}{TN + FP} \tag{11}$$

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4.1.4 Error rate (ER)

Error rate is a crucial component in assessing the proposed model's effectiveness. Here, K-fold validation is used to analyse the classification results, and the average error rate across all three classification levels is computed as,

4.1.5 G-mean

The recall rate or SN rate of each of the four classes in the final classification is shown, and the G-mean rate for the proposed model may be calculated using the following formulae.

$$G-Mean=\sqrt{SN_{normal} \times SN_{intrusion}} \tag{13}$$

4.2 Comparative analysis

The proposed model's efficacy is also assessed in light of relevant, currently conducted IoT-related research. Based on the previously described criteria, Table 2 compares the performance of the proposed model to those of comparable intelligent models studied in the literature, such as FogFL and NFL. The performance analysis demonstrates that the proposed approach outperforms the other approaches on the list. The experiment's findings demonstrate that deep learning is the foundation of the proposed strategy. The primary advantage of the proposed methodology is in its ability to promptly identify and categorise threats in highly dynamic and diverse Internet of Things networks, owing to the system's straightforward architecture and ease of implementation.

A graph that compares the results of the sensitivity rate assessment with the current model is shown in Fig. 3. For the assessments, training instances with the numbers 100, 200, 300, 400, 500, and 600 are used. The proposed model processes with a sensitivity of 88.2 percent when the matching number of training samples is used. The results demonstrate that the proposed model performs better in terms of sensitivity than competing models.Similar to the sensitivity rate analysis, the specificity rate analysis is another important assessment element in a classification model. Figure 4 illustrates the graph used in this instance. With a specificity rating of 93.85%, the results demonstrate that the proposed technique performs better than any other model in the comparison. Figure 5 displays the findings for categorization accuracy. Figure 6 displays the G-mean rate findings for assault detection. When deep learning techniques are used correctly, the model performs better. This is shown by the model's 93.87 percent classification accuracy, which is greater than that of other similar classification models used in attack detection. The accompanying image and data demonstrate how the model effectively increases the accuracy rate by using three categorization levels.We compared models and tested our framework's computation overhead. The cryptosystem's key length is fixed at 1024 bits. We consider the FNs to be the data owners and the SDP to be the model owner. Since we have clear benefits when it comes to safe machine learning training, it is noted that the suggested model has less overhead than comparable models and the results are given in Fig. 7.

Because of the significant influence on IoT data security, the attack detection model's error rate should be as low as feasible. When completing the assessments, this problem is taken into account, and Fig. 8 displays the outcomes based on the information in Table 2. The proposed model performs better than other models in terms of attack detection error rate, as the graph demonstrates. Time spent processing is another important factor. Figure 9 displays a comparison of processing time graphs. The aforementioned graphs and tables demonstrate the evident performance advantage of the

Table 2 Results for Evaluation metrics comparisons

Models	100	200	300	400	500	600
FogFL	65.1	65.1	78.1	74.9	79.7	86.1
NFL	78.2	75.2	80.0	82.2	81.7	84.3
APP	82.5	82.6	90.3	88.1	92.9	93.7

(A) Sensitivity Rate

Models	100	200	300	400	500	600
FogFL	57.2	79.2	73.4	77.2	80.3	83.1
NFL	74.5	84.6	80.9	82.8	92.3	89.5
APP	92.9	91.2	90.6	91.8	96.1	95.6

(D) G-Mean Rate

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Models	100	200	300	400	500	600
FogFL	62.1	69.7	72.2	72.8	75.5	70.9
NFL	79.5	73.9	65.4	84.3	80.5	79.2
APP	92.8	91.5	93.6	94.1	93.7	92.8

(B) Specificity Rate

Models	100	200	300	400	500	600
FogFL	45.7	49.4	55.1	60.5	52.1	49.0
NFL	40.9	30.5	32.2	44.5	31.3	32.6
APP	25.2	19.7	20.9	23.5	17.0	24.1

(E) Sensitivity Rate

Models	100	200	300	400	500	600
FogFL	64.7	66.7	83.0	83.4	69.4	74.5
NFL	73.5	74.0	72.4	73.1	89.4	85.6
APP	92.8	93.3	91.6	91.5	91.4	95.9

(C) Classification Accuracy

Models	100	200	300	400	500	600
FogFL	40.3	30.9	49.0	49.1	54.8	68.6
NFL	31.7	22.4	37.5	36.3	54.6	49.0
APP	14.9	14.4	21.4	24.6	24.7	32.8

(F) Processing Time

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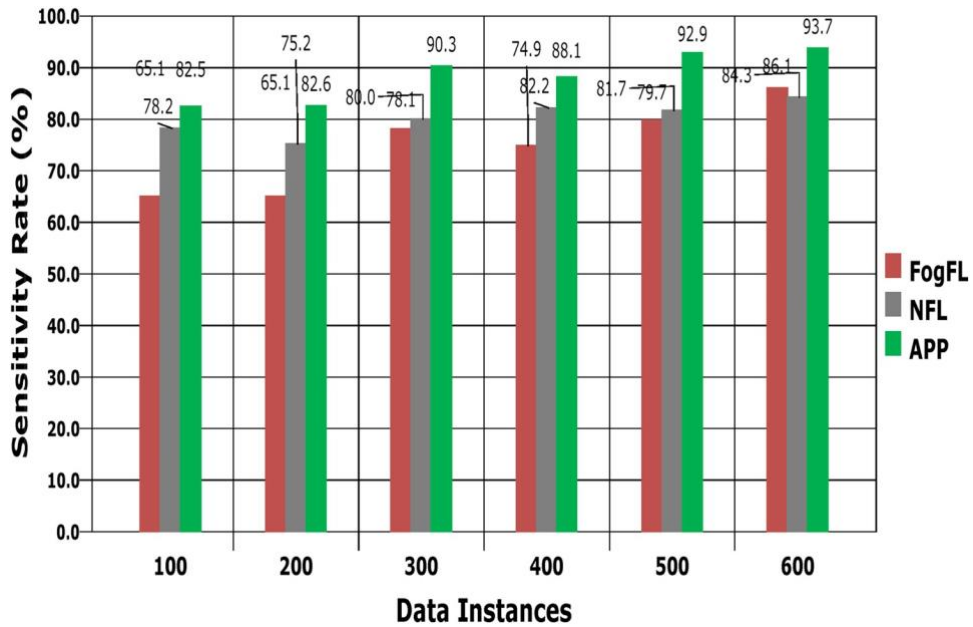

Fig. 3 Result Comparisons for Sensitivity Rate

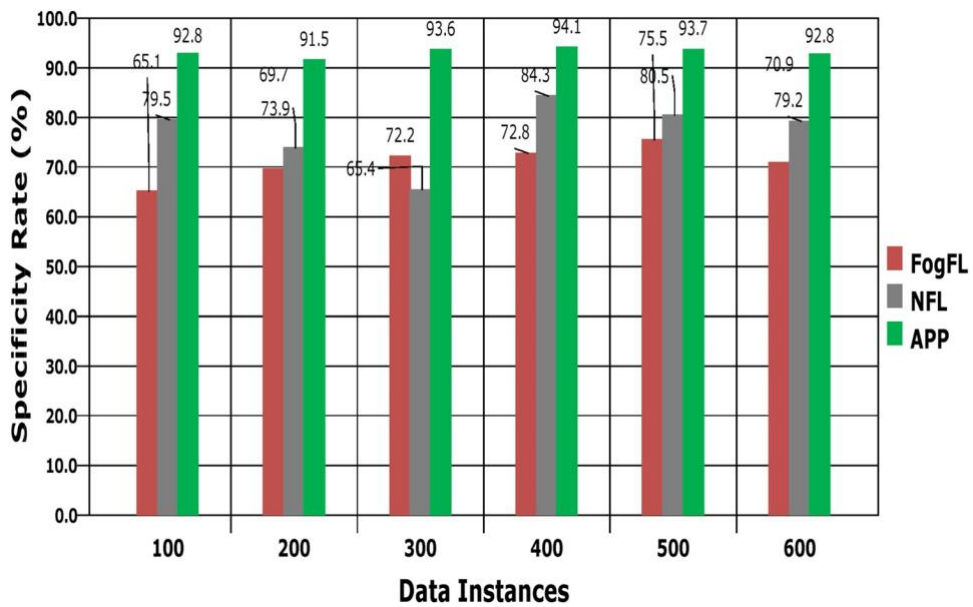

Fig. 4 Result Comparisons for Specificity Rate

proposed model, which throughout the research procedure achieved the greatest rates of classification accuracy and precision in attack detection, respectively.

5 Conclusion and future work

In this research, a machine learning framework is proposed for Advanced Privacy Protection (APP) that includes model training, model updating, and secure training data normalisation. SDP has the ability to train various machine learning models using the framework that is proposed. The accuracy of the trained models is good. The proposed approach drastically lowers the processing and enhances the data security as compared to the existing models.

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Future studies might concentrate on the privacy-preserving model's resilience and efficiency. Furthermore, sophisticated systems enable real-time data assessments. To make additional QoS prediction techniques more appropriate for edge computing, we will endeavour to enhance them inside the conventional cloud computing paradigm in

5

Classification Accuracy

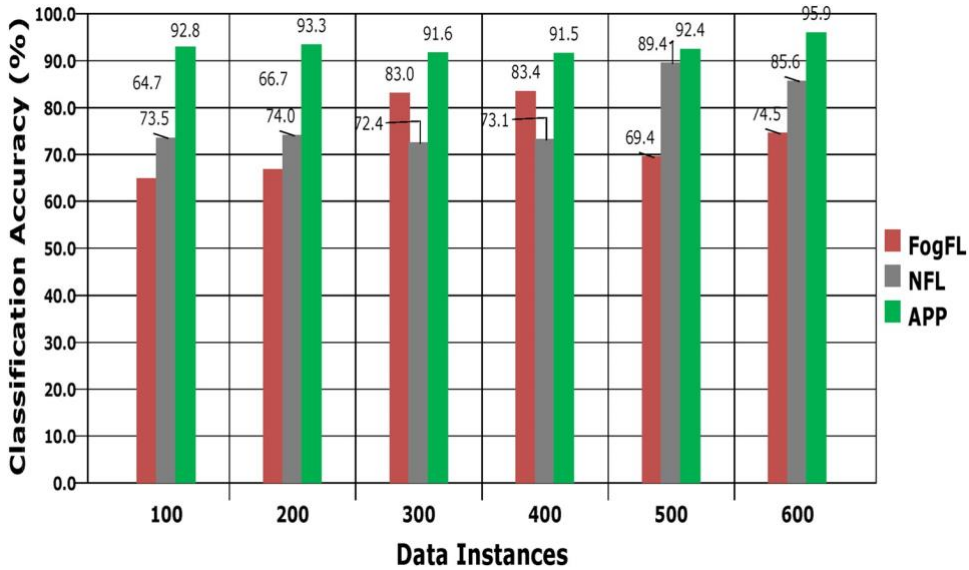


Fig. 6 Result Comparisons for G-Mean Observations

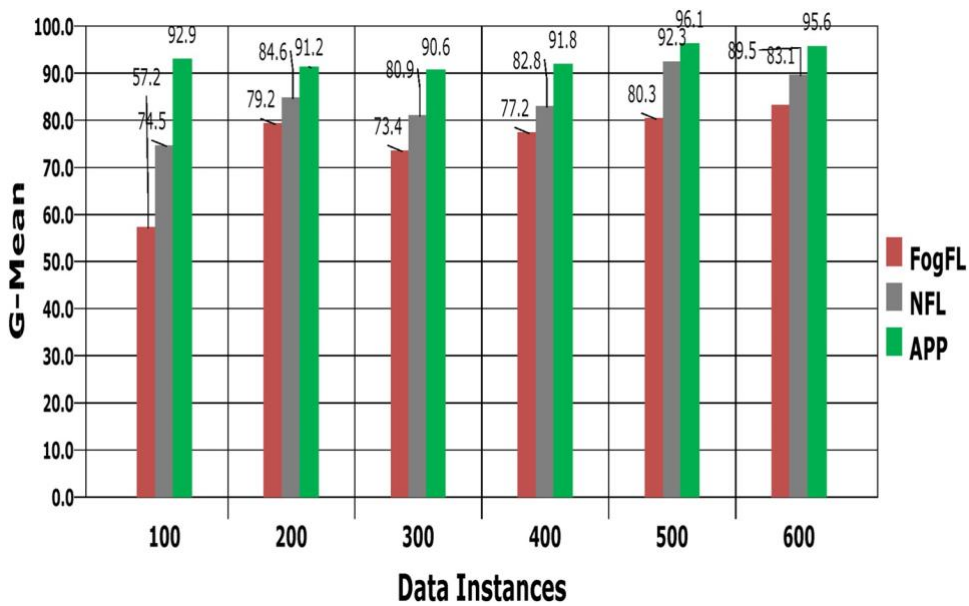
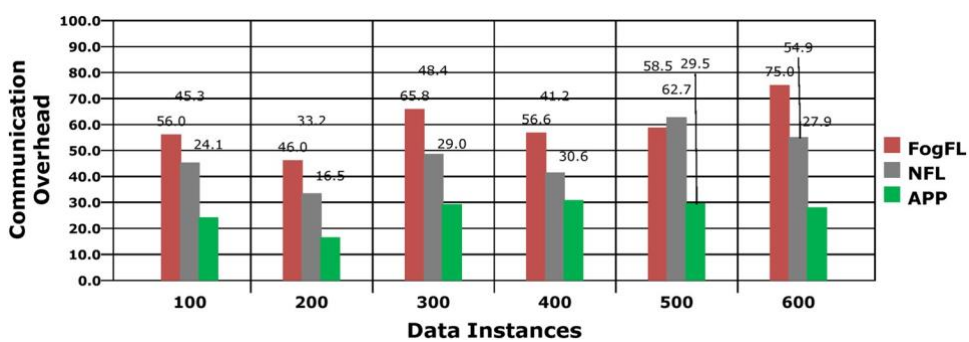


Fig. 7 Communication Overhead Analysis



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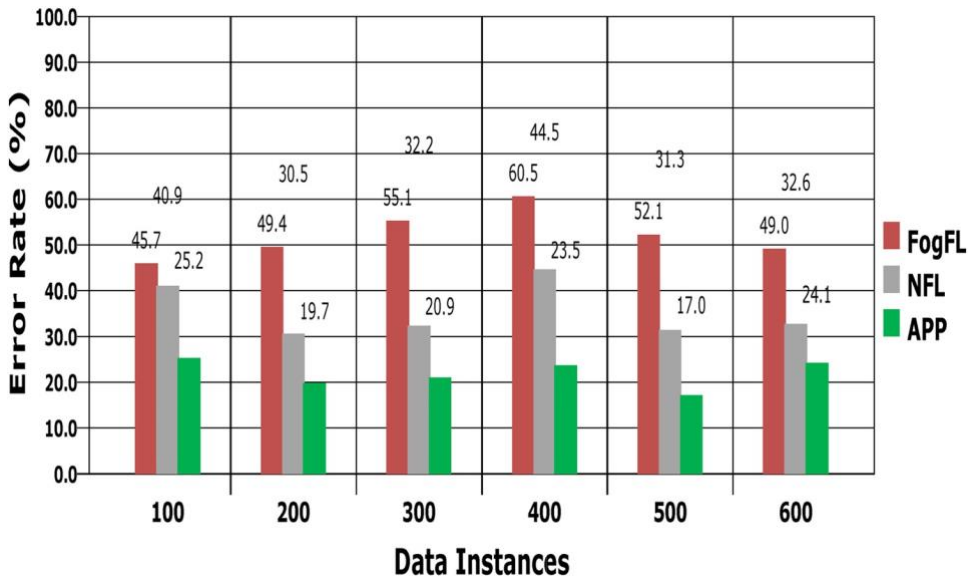

Fig. 8 Result Comparisons for Error Rate

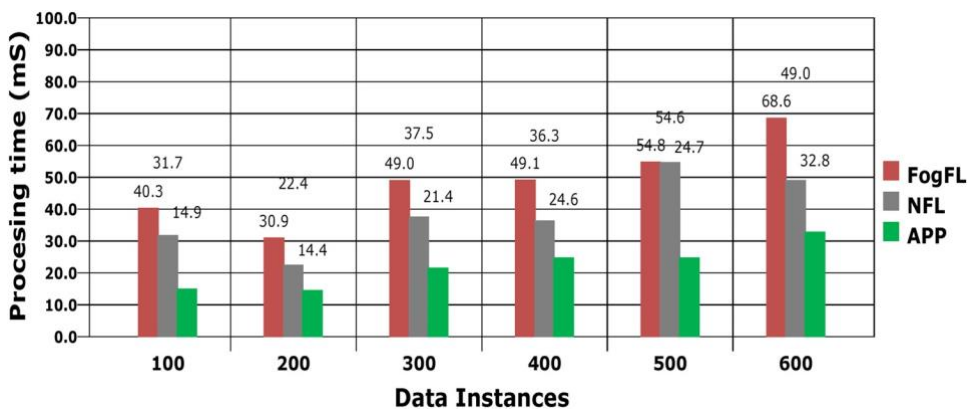

Fig. 9 Result Comparisons for Processing Time

subsequent projects. To get a much better balance between accuracy and privacy, we will also investigate alternative privacy-preserving methods for predicting edge service quality. In terms of big data processing, the research can be prolonged by focusing on data analytics along with standard key generation model.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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